

Problem Statement Worksheet (Hypothesis Formation)

What is the best model to automate white blood cell identification for clinical analysis?

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Context

Medical laboratories can have thousands of samples a day. If cell identification can be automated, it would save hours of work resulting in faster specimen turnaround times. This allows doctors and patients to get results and diagnoses faster. Also labs can continue to function with current staffing levels.

2

Criteria for success

Creating a model that accurately sorts white blood cells into neutrophils, eosinophils, basophils, monocytes, or lymphocytes.

3

Scope of solution space

Help laboratories process more manual cbc specimens daily while lowering FTEs needed for staffing. Increasing specimen turnaround times, and patient diagnosis.

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Constraints within solution space

The data set contains older movies and not movies from streaming profits

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Stakeholders to provide key insight

Clinical labs
Doctors
Patients
Healthcare professionals

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Key data sources

Microscopic cell photograph datasets from Kaggle

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